A+ LAB SERIES V5

Lab 19: Configuring Security on SOHO Networks

Document Version: 2026-01-02

Material in this Lab Aligns to the Following	
CompTIA A+ (220-1201) Exam Objectives	4.1: Explain virtualization concepts. 2.6: Given a scenario, configure basic wired/wireless small office/home office (SOHO) networks.
CompTIA A+ (220-1202) Exam Objectives	2.10: Given a scenario, apply security settings on SOHO wireless and wired networks.

Contents

Introduction

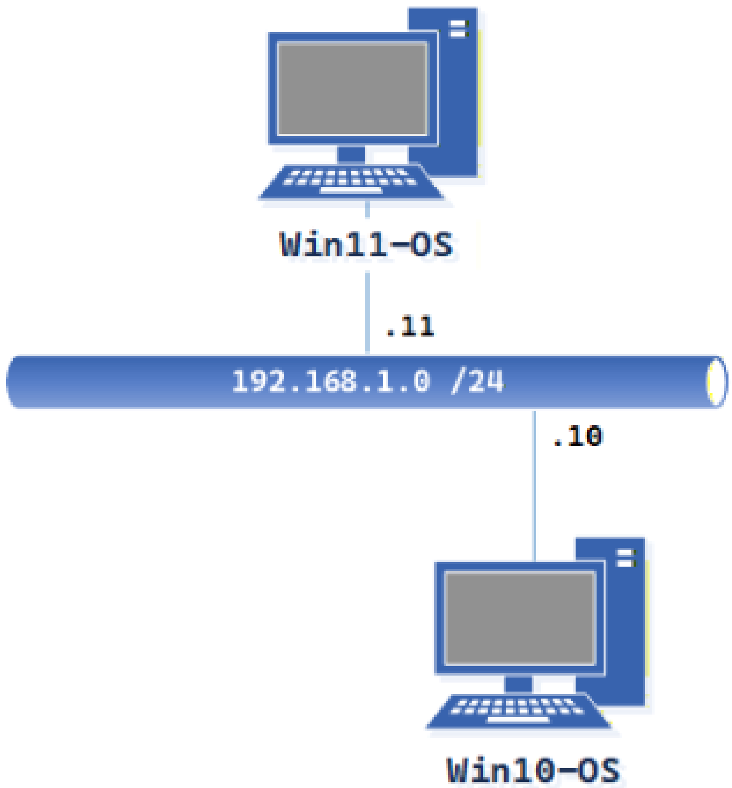
In this lab, you'll configure appropriate security settings on small office/home office (SOHO) wireless networks. In the realm of modern computing, the security of home and small office networks is paramount. With the increasing reliance on internet-connected devices and the growing sophistication of cyber threats, it's imperative to establish robust security measures.

Objective

In this lab, you'll configure appropriate security settings on small office/home office (SOHO) wireless networks.

- Windows 11 Hyper-V Setup and Availability
- Configure Firewall settings to meet lab requirements.
- Changing Default Credentials.
- Assigning static IP addresses.
- Enabling MAC filtering.
- Enabling Port filtering.
- Disabling unused ports.
- Set Content filtering.

Lab Topology



Lab Settings

The information in the table below will be needed to complete the lab. The task sections below provide details on the use of this information.

Virtual Machine	IP Address	Account (if needed)	Password (if needed)
Win11-OS	192.168.1.11 / 24	`sysadmin`	`NDGlabpass123!`
Win10-OS	192.168.1.10 / 24	`sysadmin`	`NDGlabpass123!`

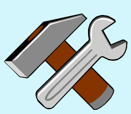
1 Windows Hyper-V Setup and Availability

Windows 11 Hyper-V is a robust virtualization platform built into the operating system, enabling you to create and manage virtual machines (VMs) on your Windows 11 device. This powerful tool offers a wide range of capabilities for various use cases. The key features and benefits include:

- Create and Manage Virtual Machines: Easily create virtual machines running different operating systems, such as Windows, Linux, or other supported platforms.
- Test Software in Isolated Environments: Safely test new software or updates without affecting your primary operating system.
- Run Multiple Operating Systems Simultaneously: Efficiently run multiple operating systems on a single physical machine.
- Development and Test Applications: Create isolated environments for developing and testing applications.
- Learn New Technologies: Experiment with different operating systems and software without impacting your primary system.
- Enhanced Security: Isolate potentially risky software or applications within virtual machines to protect your host system.

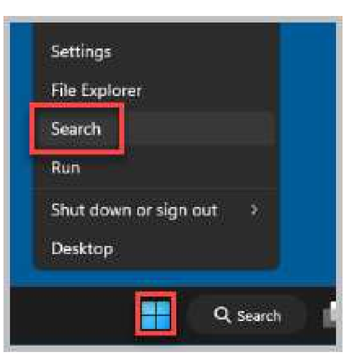
1.1 Verifying that Hyper-V is Available

1. Open a console connection to the **Win11-OS** system.

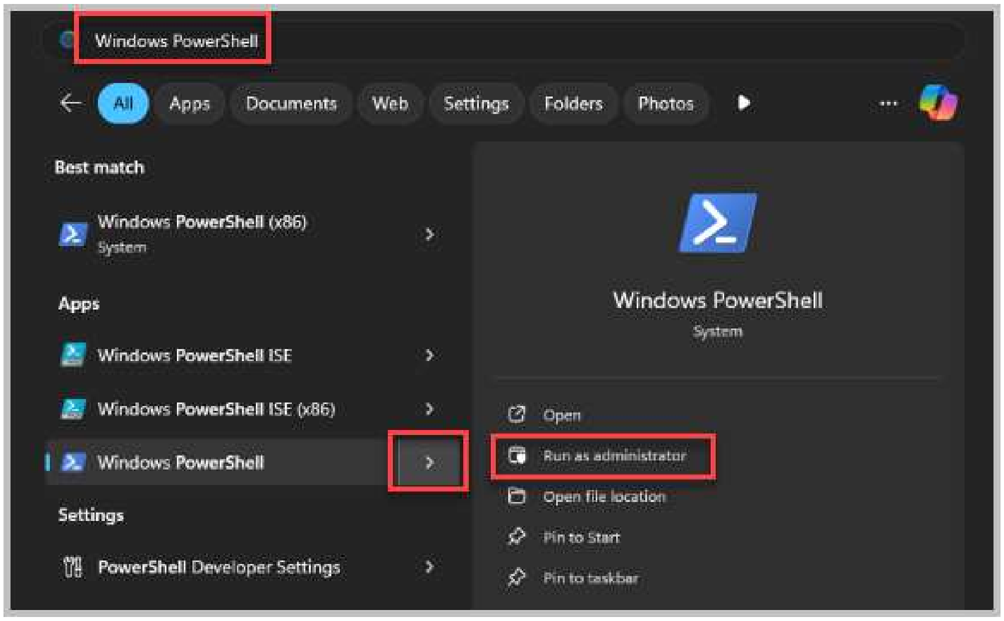


To launch the console for a lab device, you can click on the respective tab in the navigation bar at the top or navigate to the *Topology* section and click the lab device’s corresponding image.

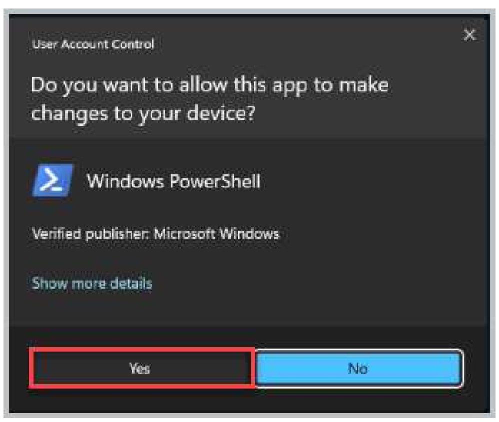
2. Right-click **Start** and select **Search.**



3. In the search text box, type **Windows PowerShell**, select the arrow next to **Windows PowerShell** from the results, and choose **Run as Administrator**.

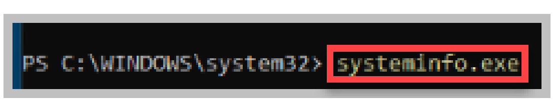


4. Click **Yes** on the *UAC* prompt.



5. At the prompt, type the command below, followed by pressing the **Enter** key.

```
systeminfo.exe
```

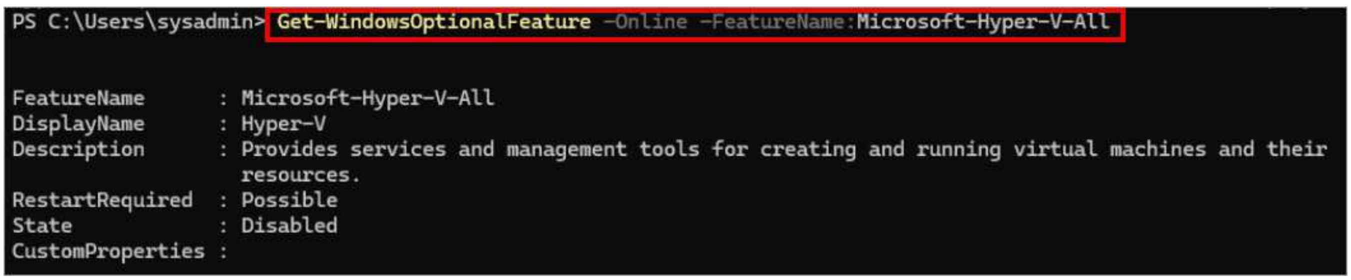


6. Scroll down to the bottom and locate the *Hyper-V Requirements* section to check if the PC can run Hyper-V. If this says "No hypervisor has been detected," the system cannot run Hyper-V. You can see on this machine that a hypervisor has been detected, indicating that the lab machine can run it.

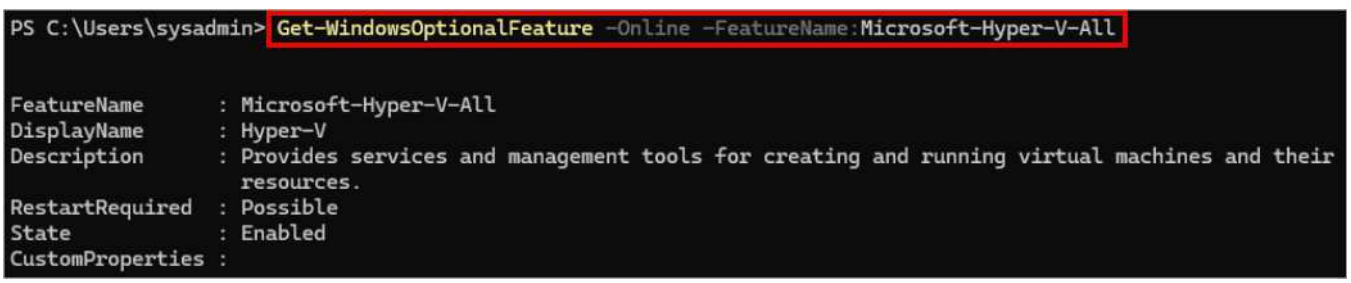
7. Check if the Hyper-V feature is enabled by entering the command below.

```
Get-WindowsOptionalFeature -Online -FeatureName:Microsoft-Hyper-V -All
```

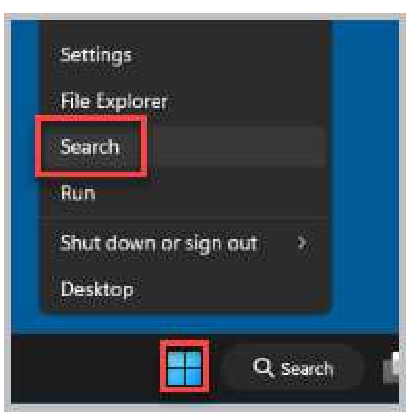
If you see output similar to the one below, it indicates that the Hyper-V feature is disabled.



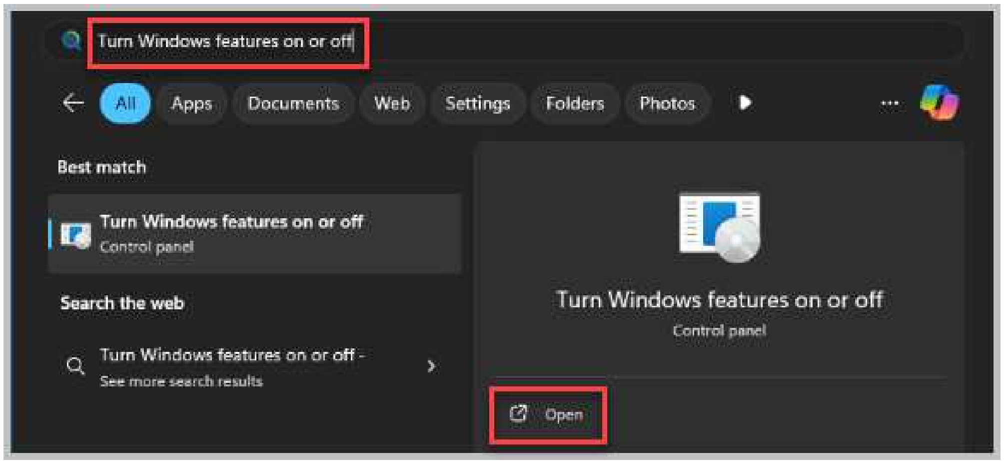
If you see output similar to the one below, it indicates that the Hyper-V feature is enabled.



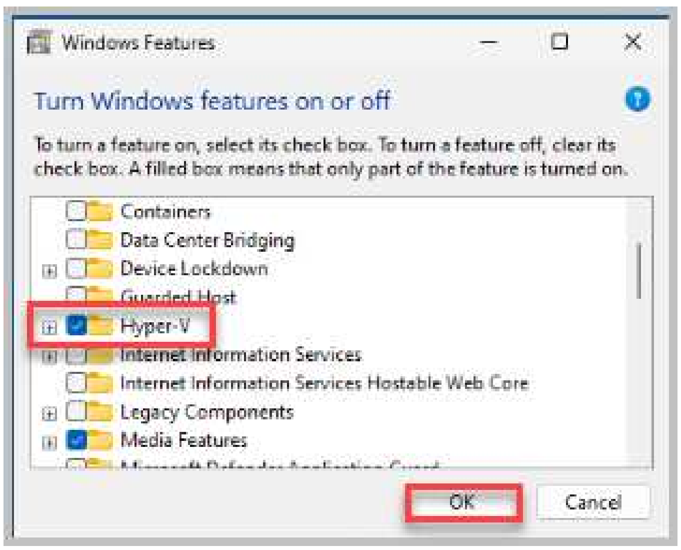
8. You can also check to see if the *Hyper-V* feature is enabled using the Graphical User Interface (GUI). Right-click the **Windows** icon on the taskbar and select **Search.**



9. Type **Turn Windows features on or off**. Select **Open.**



10. Verify that a check is in the checkbox next to *Hyper-V* in the *Windows Features* dialog box. Click **OK**.



If Hyper-V was not previously enabled, this box would not be checked, and you would need to check it and reboot the machine to apply the change.

11. Leave *Win11-OS* open and continue to the next task.

1.2 Enabling Hyper-V

If the Hyper-V feature is not enabled, follow these steps to enable it. Otherwise, skip to the next task.

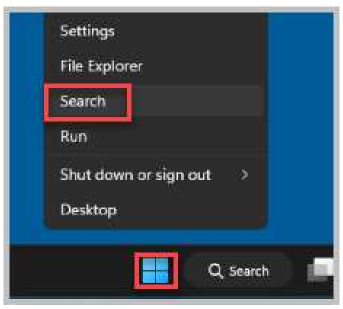
1. In the *PowerShell* prompt, enter the command below. Notice that the system needs to be restarted to take effect. Press **Y**, then press **Enter** to reboot.

```
Enable-WindowsOptionalFeature -Online -FeatureName:Microsoft-Hyper-V -All
```

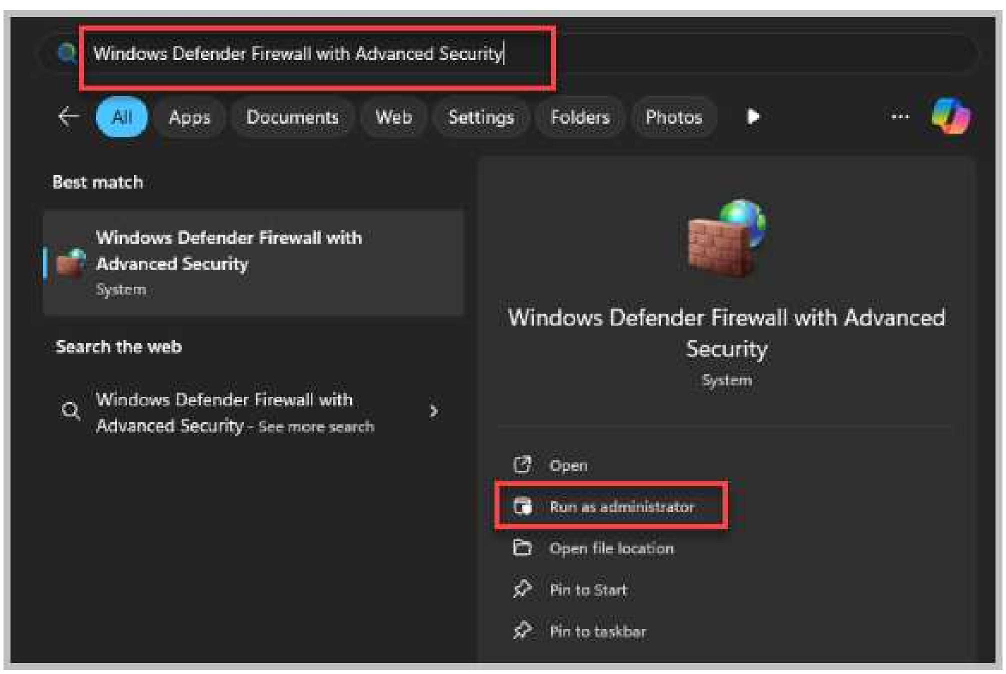
2. After the reboot, leave *Win11-OS* open for the next lab activity.

1.3 Disable Win11-OS Firewall

1. To establish connectivity between the OpenWRT VM and Win11-OS, ensure the Win11-OS Firewall is disabled. Right-click the *Win11-OS **Start Icon*** and select **Search.**

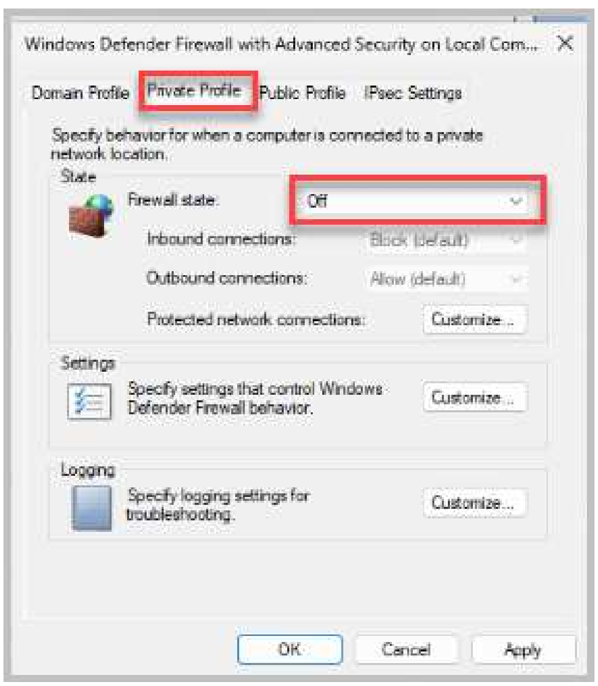


2. In the *Search for apps, settings, and documents* textbox, type **Windows Defender Firewall with Advanced Security,** and select **Run as administrator.**

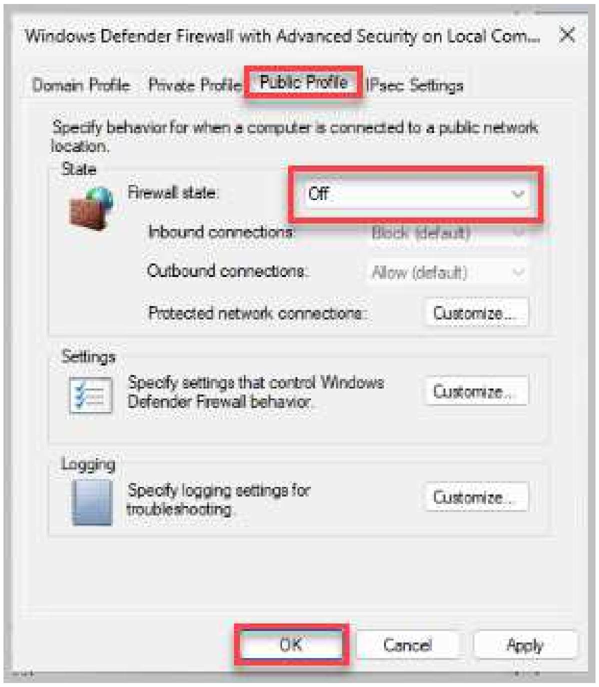


3. Select **Windows Defender Firewall Properties.**

4. Select the **Private Profile** tab, and turn the *Firewall state **Off.***



5. Select the **Public Profile** tab, and turn the *Firewall state* **Off**. Select **OK** to commit changes.



6. Close the *Windows Defender Firewall with Advanced Security* window. Leave the *Win11-OS* console open for the next activity.

2 Configuring Security on SOHO Networks

In the contemporary digital landscape, the security of home and small office (SOHO) wireless networks has become a critical concern. As the number of internet-connected devices proliferates and cyber threats become more complex, implementing robust security measures is essential to safeguard sensitive data and protect against unauthorized access. Network administrators can mitigate risks by configuring appropriate security settings and ensuring the confidentiality, integrity, and availability of network resources.

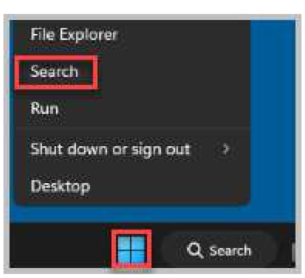
Key security measures for SOHO networks include:

- Changing Default Credentials: Modifying the default username and password for the router's administrative interface to prevent unauthorized access.
- Assigning Static IP Addresses: Assigning static IP addresses to devices on the network to enhance network management and security.
- Enabling MAC Filtering: Restricting network access to devices with authorized MAC addresses to prevent unauthorized access.
- Enabling Port Filtering: Restricting network access to devices with authorized Port Rules to prevent unauthorized access.
- Disabling Unused Ports: Disabling unnecessary ports on the router reduces potential vulnerabilities.
- Content Filtering: Implementing content filtering to block access to malicious websites and harmful content.
- Strong Password Policies: Utilizing complex, unique passwords for both the router's administrative interface and the wireless network itself.
- Firewall Configuration: Implementing firewall rules to filter incoming and outgoing traffic, blocking unnecessary ports and services.

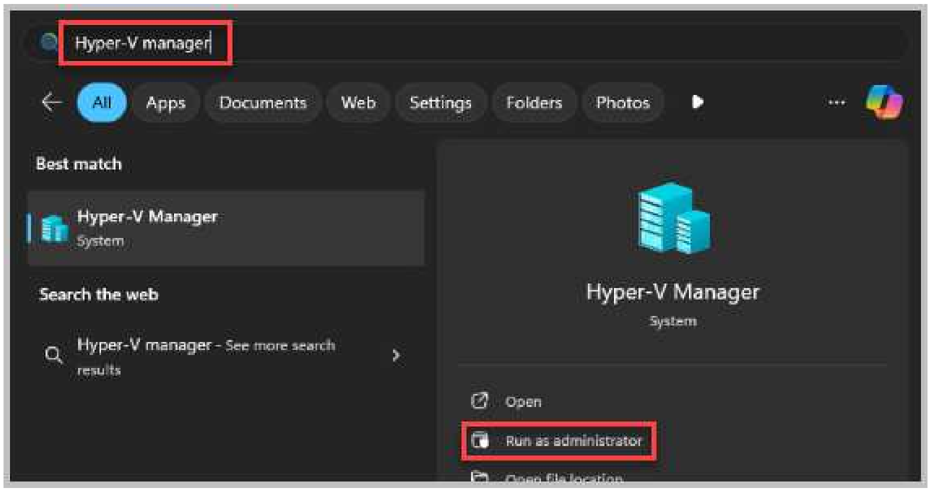
By diligently implementing these security measures, individuals and small businesses can significantly enhance the security posture of their SOHO networks, safeguarding valuable data and mitigating the risk of cyberattacks.

2.1 Configuring OpenWRT Virtual Machine for Access

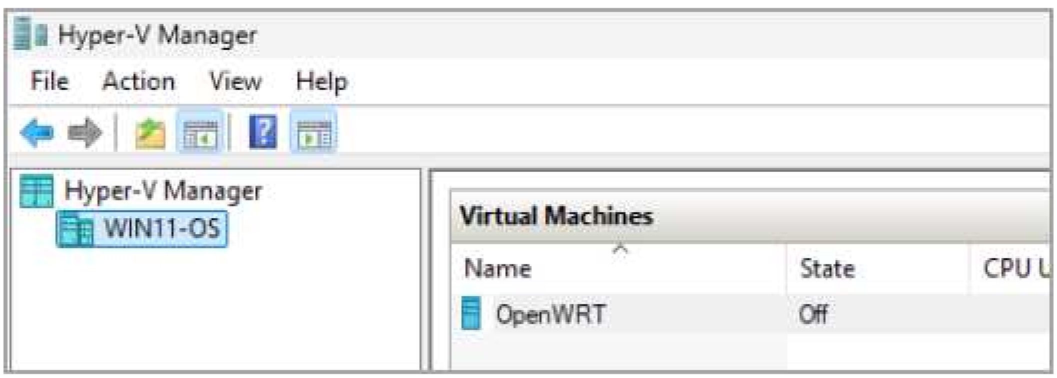
1. While on the *Win11-OS* system, right-click on the **Start** button and select **Search.**



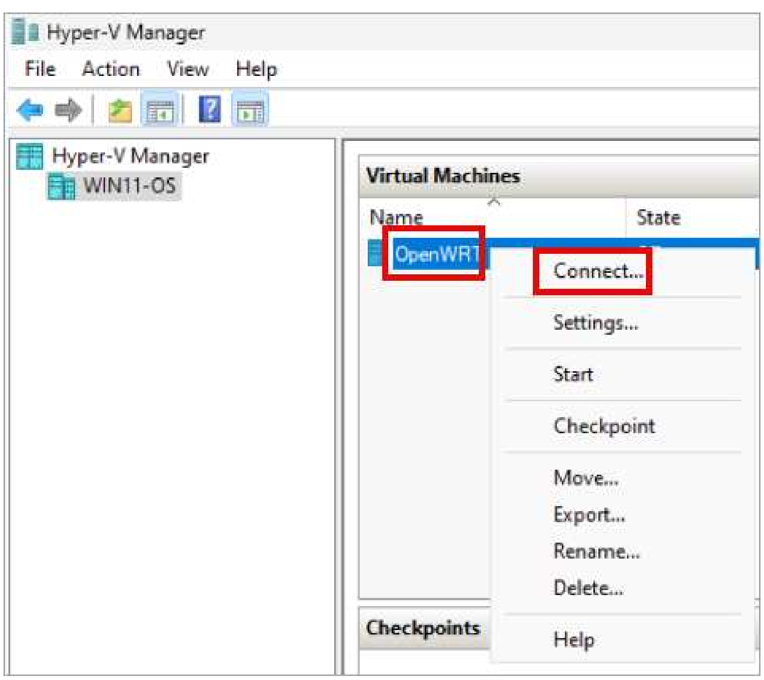
2. Type **Hyper-V Manager** and select **Run as Administrator**.



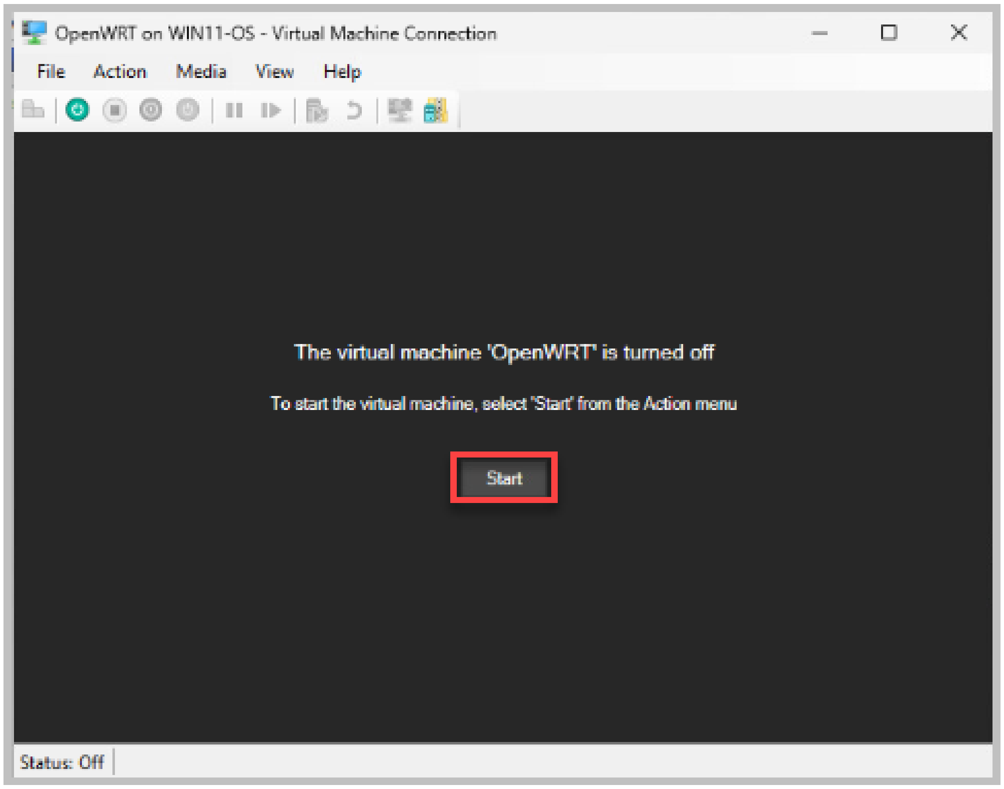
3. You will now see a list of pre-configured virtual machines with the Hyper-V Manager. Notice the OpenWRT VM is in the *Off* State. The VM is not powered on.



4. Right-click the **OpenWRT** virtual machine and select **Connect.**



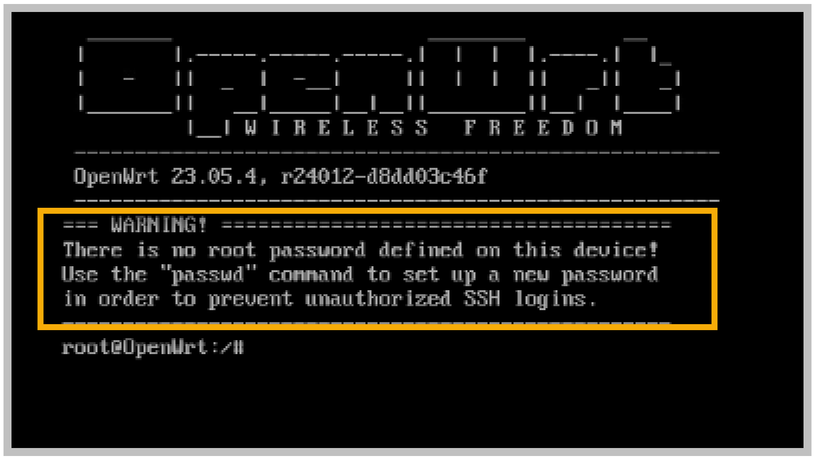
5. Select **Start** for the *OpenWRT* Virtual Machine (VM).



6. Expand the *OpenWRT* window to make it easier to complete the requirements of this lab.

7. Select **Enter** on your keyboard to activate the *OpenWRT Console.*

8. As this is a new OpenWRT installation, no root password is defined. Once the OpenWRT Virtual Machine successfully boots, your console should list a warning that the *root* password has not been configured for this device.



OpenWRT is a Linux-based operating system specifically designed for embedded devices, primarily used on routers and other network devices. It offers a highly customizable and open-source platform that allows users to tailor their devices to their specific needs.

Key Features of OpenWRT:

- **Fully Customizable:** OpenWRT provides a fully writable filesystem and package management system, enabling users to install and configure a wide range of software packages.
- **Open-Source:** As an open-source project, OpenWRT is freely available and allows for community contributions and customization.
- **Versatile:** It can be used for various applications, including routing, VPN, firewalling, and more.
- **Community-Driven:** A large and active community provides support, documentation, and custom firmware builds.

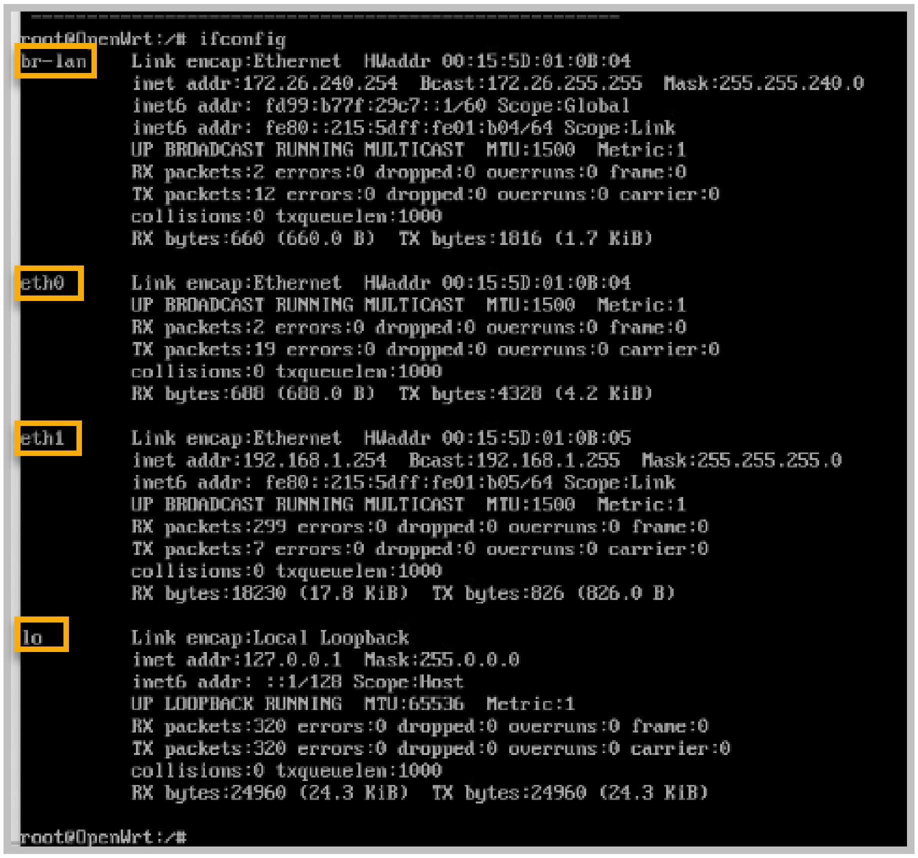
9. Leave the console connection open for the next lab activity.

2.2 Verifying OpenWRT Static IP Address

1. There may be an error where everything you type is capitalized, but the Caps Lock on your keyboard is not activated. You may need to press the Caps Lock key on your keyboard, type reboot, and press Enter.

2. In the OpenWRT console, enter the command **`ifconfig`.**

3. The output from the *ifconfig* command lists four network interfaces: *br-lan, eth0, eth1, and lo.*



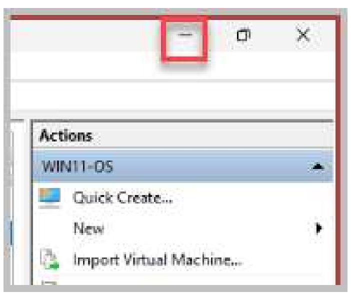
4. The IP Address has been pre-configured for network interface *eth1,* which is listed as 192.168.1.254 with a subnet address of 255.255.255.0. This is the address you will use to access the OpenWRT web interface.
5.

Understanding 192.168.1.254/255.255.255.0
IP Address: 192.168.1.254 **Subnet Mask:** 255.255.255.0
Breaking Down the IP Address

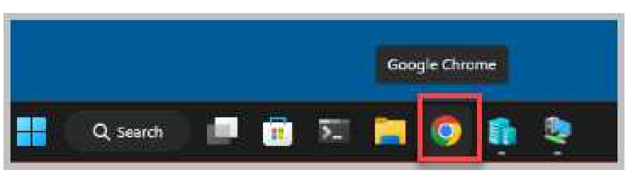
- **192.168.1.254:** This is a private IP address, meaning it's not routable on the public internet. It's commonly used for home and small office networks.
- **255.255.255.0:** This is the subnet mask, which defines the network and host portions of the IP address. In this case, it indicates a network with a maximum of 254 hosts.

6. Minimize the *OpenWRT VM* window.

7. Minimize the *Win11-OS Hyper-V Manager* window.

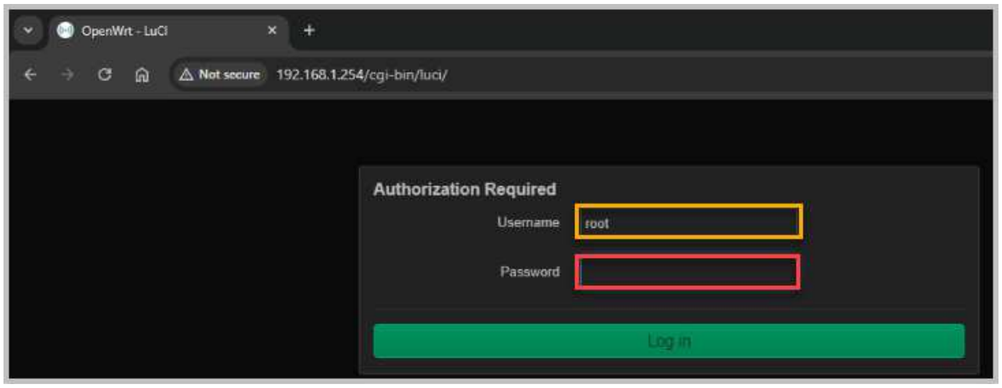


8. Select the **Google Chrome** icon on the taskbar of the *Win11-OS System.* Maximize the *Google Chrome Window.*



9. In the address textbox in *Google Chrome,* type `192.168.1.254` and select **Enter** on your keyboard. This opens the web interface of *OpenWRT,* which is called *LuCI.*

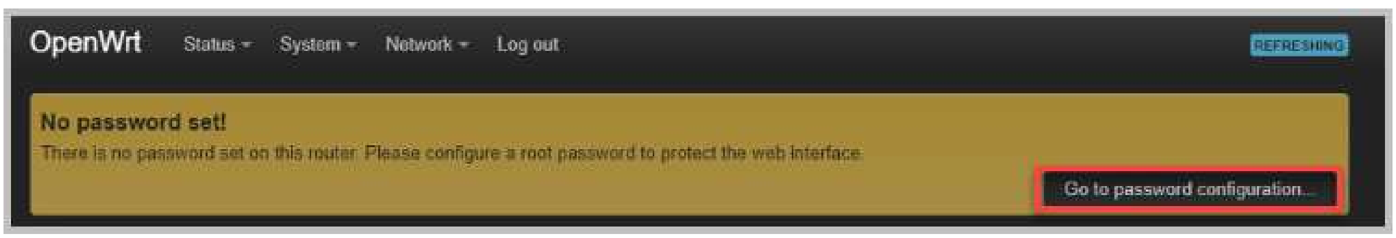
10. To access the *OpenWRT LuCI* web interface, you'll need to provide the correct credentials. Since the default username is *root* and no password has been set, you can simply log in by clicking the **Log in** button.



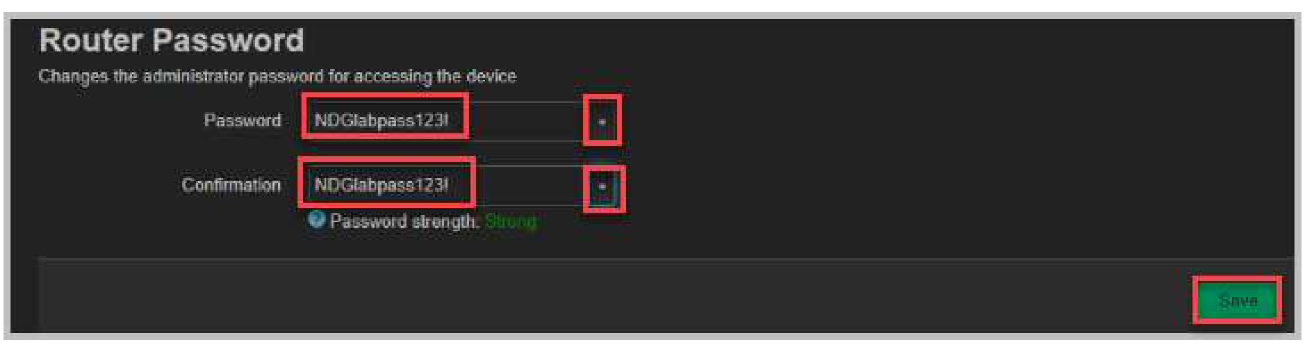
11. Leave the *OpenWRT* window open for the next lab activity.

2.3 Changing Default Credentials

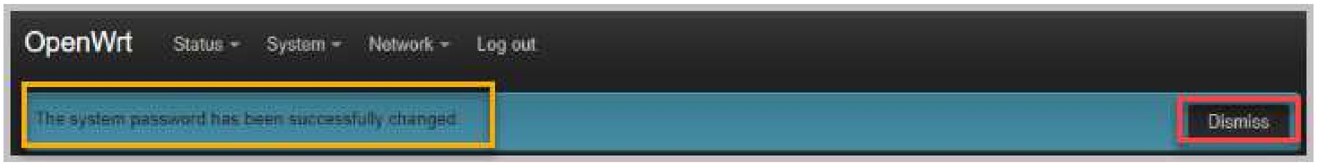
1. As there is no password configured for the *root account,* select **Go to password configuration** in the *OpenWRT* window.

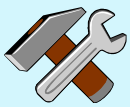


2. Type `NDGlabpass123!` for both the *Password and Confirmation***.** To see the password you are typing, select the **Star** next to each password you entered. Select **Save.**



3. You should see a notification that the system password has been successfully changed. Select **Dismiss** to acknowledge the notification.





Security Best Practices:

- **Strong Password:** Use a strong, unique password that combines uppercase and lowercase letters, numbers, and symbols.
- **Regular Password Changes:** Regularly change your password to enhance security.
- **Enable SSH Key-Based Authentication:** Consider using SSH keys for a more secure authentication method.
- **Firewall Configuration:** Configure your firewall to restrict access to your device.
- **Keep Software Updated:** Regularly update your OpenWRT firmware and packages to address security vulnerabilities.

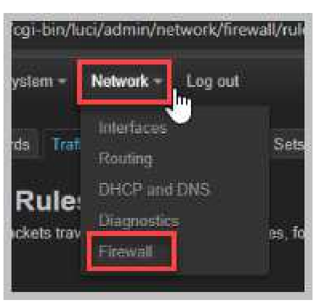
By following these steps and adhering to security best practices, you can ensure the security of your OpenWRT device.

4. Leave the *OpenWRT* window open for the next lab activity.

2.4 Enabling MAC Filtering

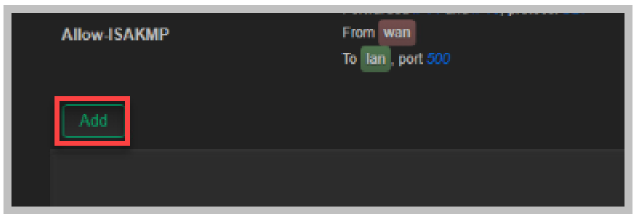
MAC filtering is a security feature that allows you to restrict network access to devices with specific MAC addresses. This can help prevent unauthorized access to your network.

1. In the *OpenWRT LuCI* web interface, select **Network** and then **Firewall.**

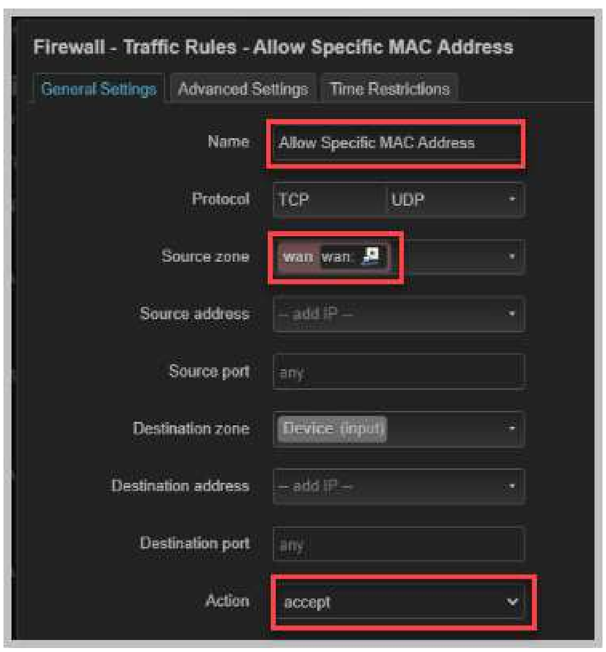


2. Select the **Traffic Rules** tab.

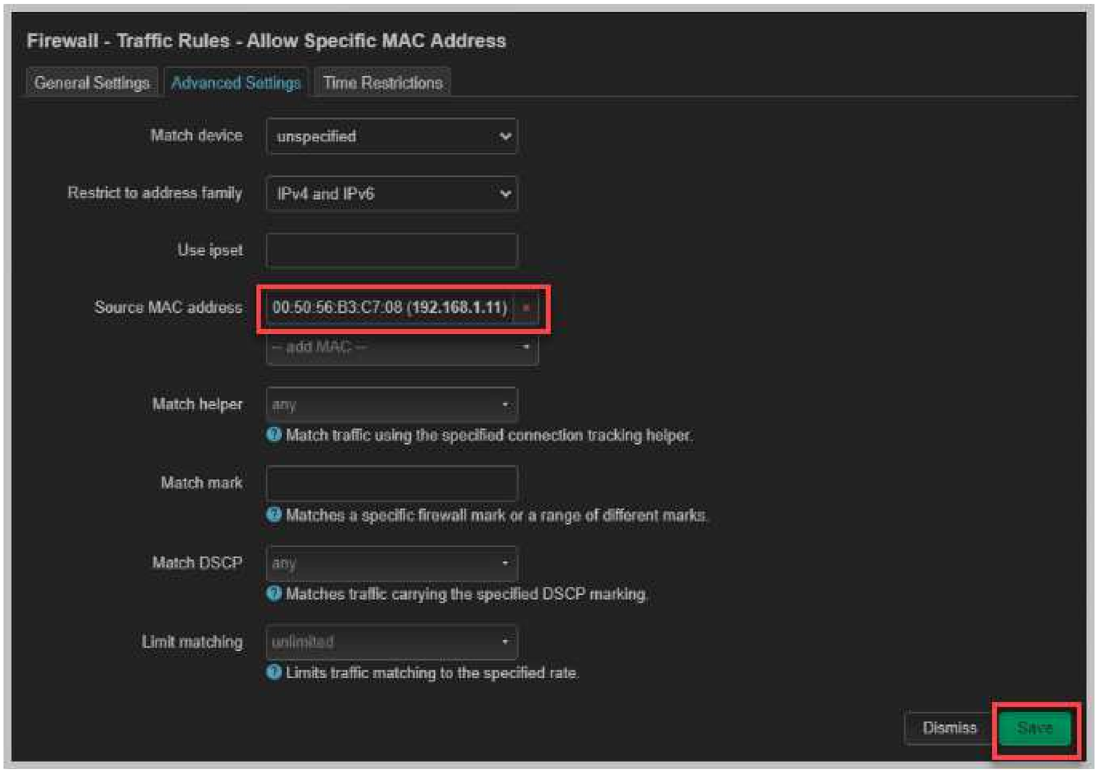
3. Scroll to the bottom of the *Traffic Rules Page,* and select **Add.**



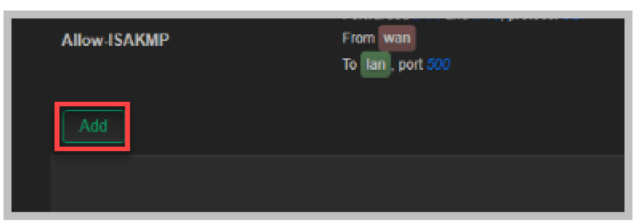
4. On the **General Settings** Tab, type **Allow Specific MAC Address** in the Name textbox, and change the Source zone to **wan.** This sets a name for the Traffic Rule and isolates the WAN Zone Traffic by selecting the wan Source zone. For action, select **accept.** This sets the Firewall Traffic Rule to accept only Specific Mac Addresses.



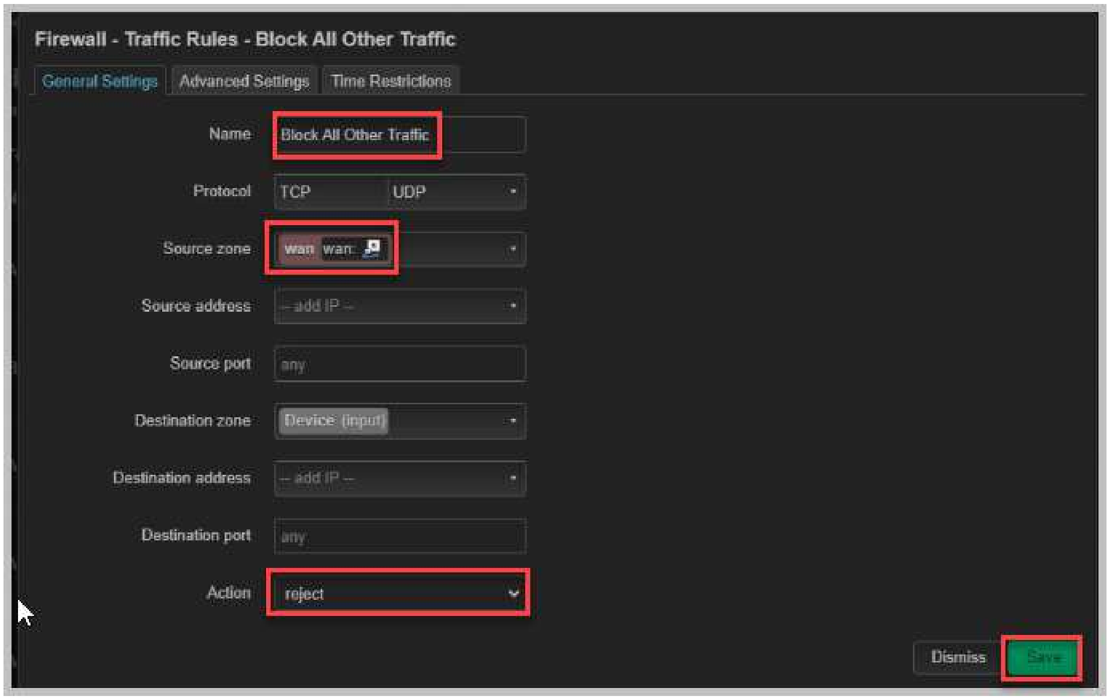
5. Select the **Advanced Settings** tab. Select the drop-down arrow for **Source Mac Address,** then select the Windows 11 IP Address (192.168.1.11). This selection also lists the *MAC address* associated with the Windows 11 Network Interface: **00.50.56.b3.c7.08.** Select **Save.**



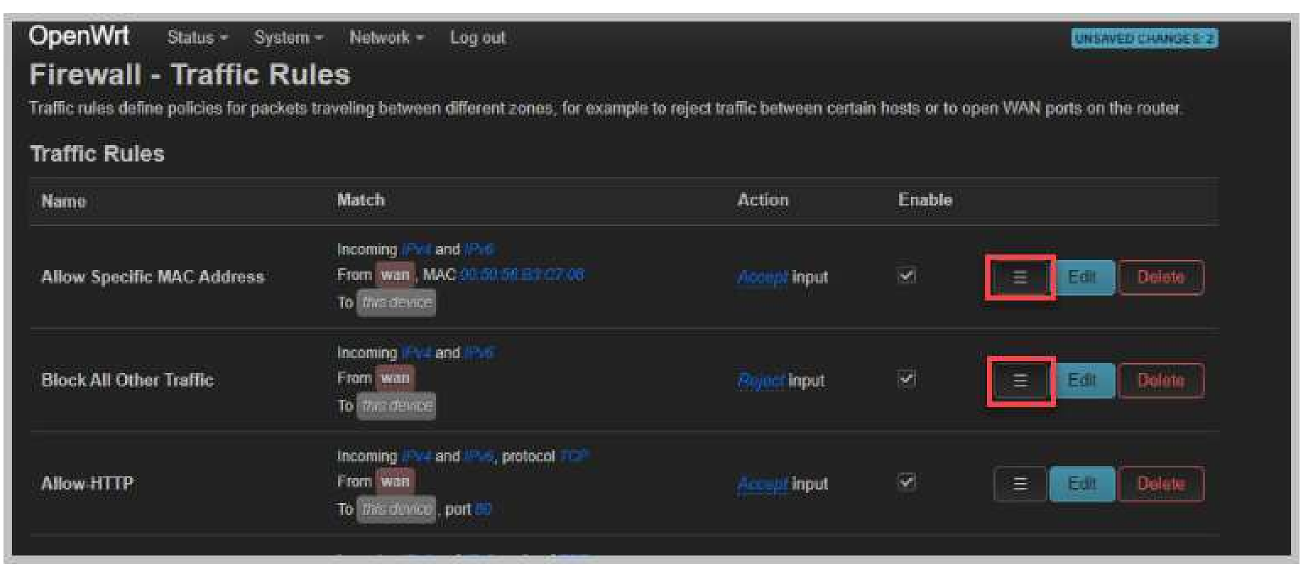
6. At the bottom of the *Traffic Rules Page,* select **Add.**



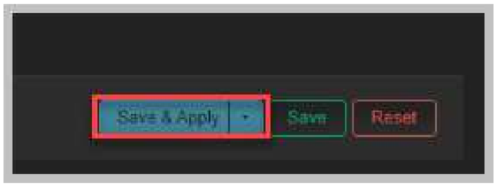
7. On the **General Settings** Tab, type `Block All Other Traffic` in the Name textbox, and change the Source zone to **Wan.** This sets a name for the Traffic Rule, and by selecting the Wan Source zone, this isolates the WAN Zone Traffic. For action, select **reject.** This sets the Firewall Traffic Rule to reject all other traffic from the *WAN Zone.* Select **Save.**



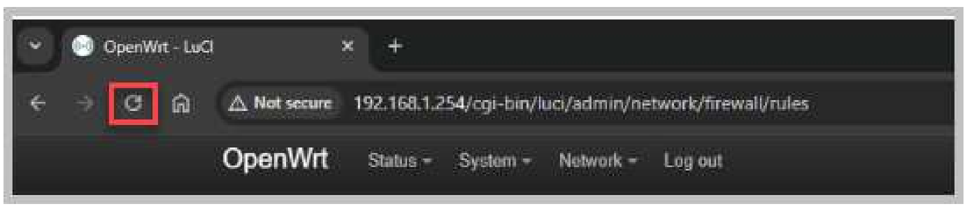
8. To ensure that the new rules are processed first and take precedence over other rules, you'll need to move them to the top of the traffic rules list. By moving the new rules to the top of the list, you ensure they are processed first by the firewall, effectively blocking all other traffic except for the specified MAC address.
9. Move both new Traffic Rules to the beginning of the Traffic Rules List. First, move the **Allow Specific Mac Address** rule by selecting and holding down the **Drag to reorder** button and dragging it to the top of the list. Move the **Block All Other Traffic** rule below the *Allow Specific Mac Address rule.* Your Traffic Rules list should look exactly like the screenshot below.



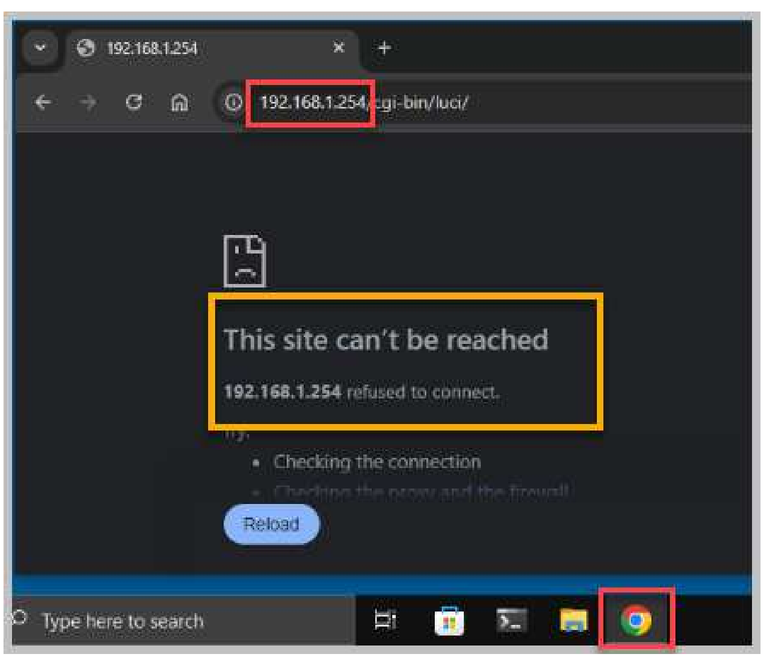
10. Scroll to the bottom of the *Traffic Rules* list, and click the **Save & Apply** button. This action will save your changes to the device's storage and immediately apply them to the active configuration. This ensures that the newly configured rules are enforced by the firewall.



11. To verify that the new traffic rules are working correctly, refresh the OpenWRT LUCI webpage. If the page refreshes successfully, it indicates that the first firewall rule, which allows only specific MAC addresses to connect, is working as intended.



12. Open a console connection to the **Win10-OS** system.
13. Select the **Google Chrome** icon on the taskbar. Enter **192.168.1.254** in the address window and select **Enter** on your keyboard. You should get a notification that *This site can't be reached.* This indicates that the second firewall rule, which blocks all other traffic to connect, is working as intended.



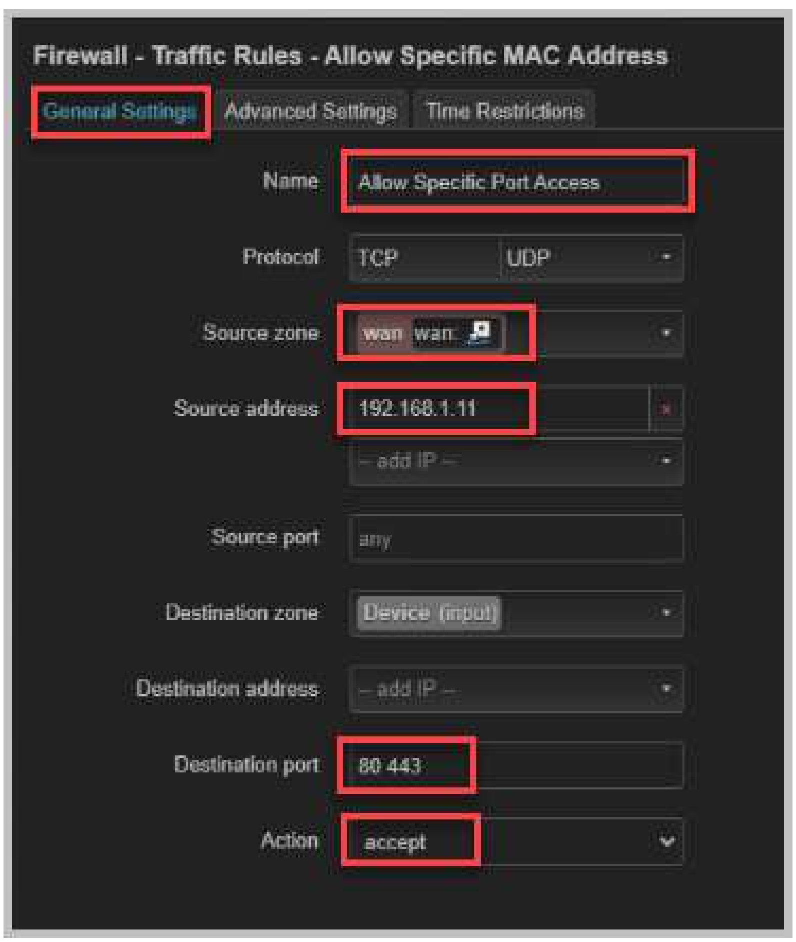
14. Leave the *Win11-OS OpenWRT* window open for the next lab activity.

2.5 Enabling Port Filtering

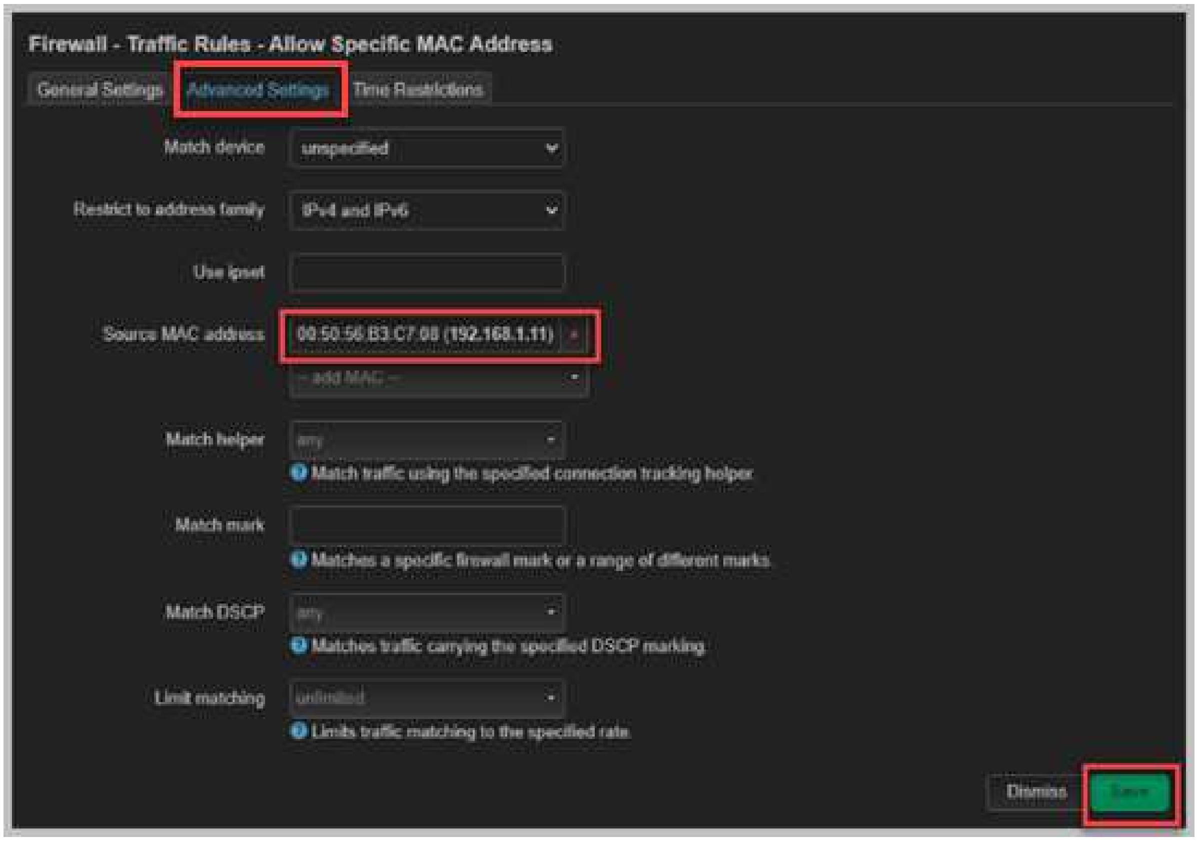
Port filtering is a security feature that allows you to restrict network access to devices with specific Ports. This can help prevent unauthorized access to your network.

1. Select **Edit** for the **Allow Specific MAC Address** Traffic Rule.

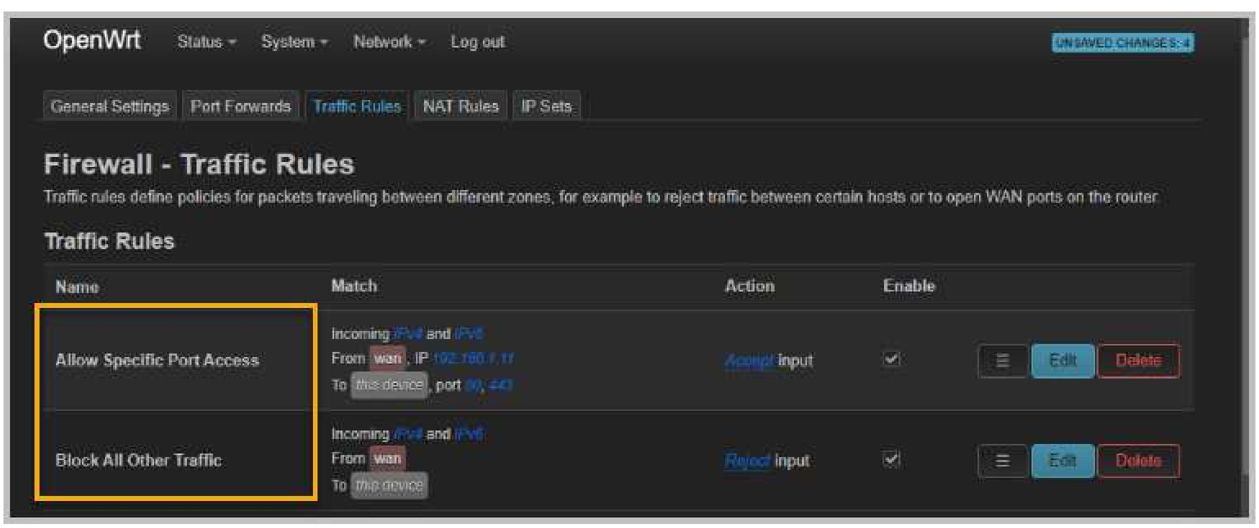
2. On the **General Settings** Tab, change the name to **Allow Specific Port Access**, and verify that the Source zone is set to **WAN.** Select the drop-down arrow for *Source* Address and select **192.168.1.11,** which is the IP Address of the *Win11-OS System.* In the *Destination* Port, type **80 443.** This sets a name for the Traffic Rule and isolates the WAN Zone Traffic by selecting the Wan Source zone. This also isolates that the *Win11-OS System* is the only computer accessing *OpenWRT* via ports 80 or 443. For action, select **accept.** This sets the Firewall Traffic Rule to accept only Specific Ports for access.



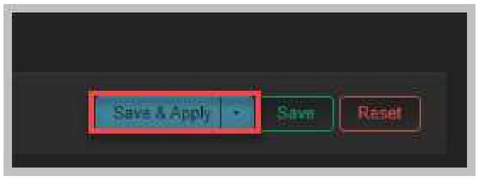
3. Select the **Advanced Settings** tab. Select the **X** for *Source MAC Address* to remove the **192.168.1.11** address from the list. Select **Save.**



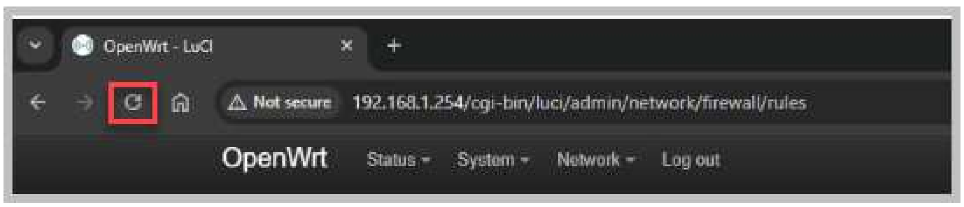
4. Verify that the **Allow Specific Port Access** is listed first and the **Block All Other Traffic** is listed second in the list of *Firewall Traffic Rules.* Your Traffic Rules list should look exactly like the screenshot below.



5. Scroll to the bottom of the *Traffic Rules* list, and click the **Save & Apply** button. This action will save your changes to the device's storage and immediately apply them to the active configuration. This ensures that the newly configured rules are enforced by the firewall.

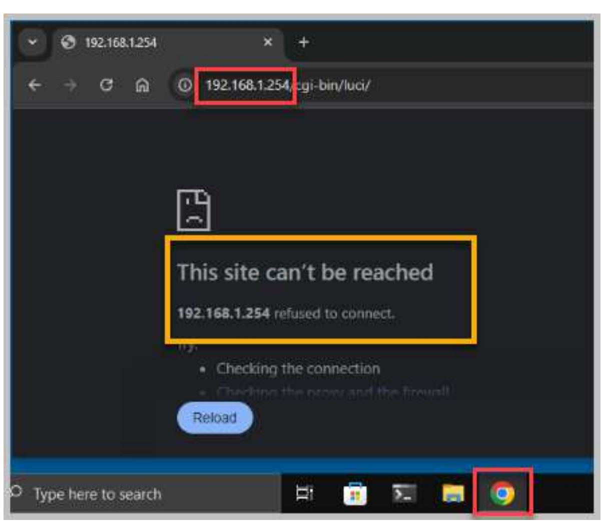


6. To verify that the new traffic rules are working correctly, refresh the OpenWRT LUCI webpage. If the page refreshes successfully, it indicates that the first firewall rule, which allows only specific MAC addresses to connect, is working as intended.



7. Open a console connection to the **Win10-OS** system.

8. Select the **Google Chrome** icon on the taskbar. Enter **192.168.1.254** in the address window and select **Enter** on your keyboard. You should get a notification that *This site can't be reached.* This indicates that the second firewall rule, which blocks all other traffic to connect, is working as intended.



9. Open the console connection back to the **Win11-OS** system.

10. Delete the first and second rule you created under **Network>Firewall>Traffic Rules.**

11. Scroll to the bottom of the *Traffic Rules* list and select **Save & Apply.**

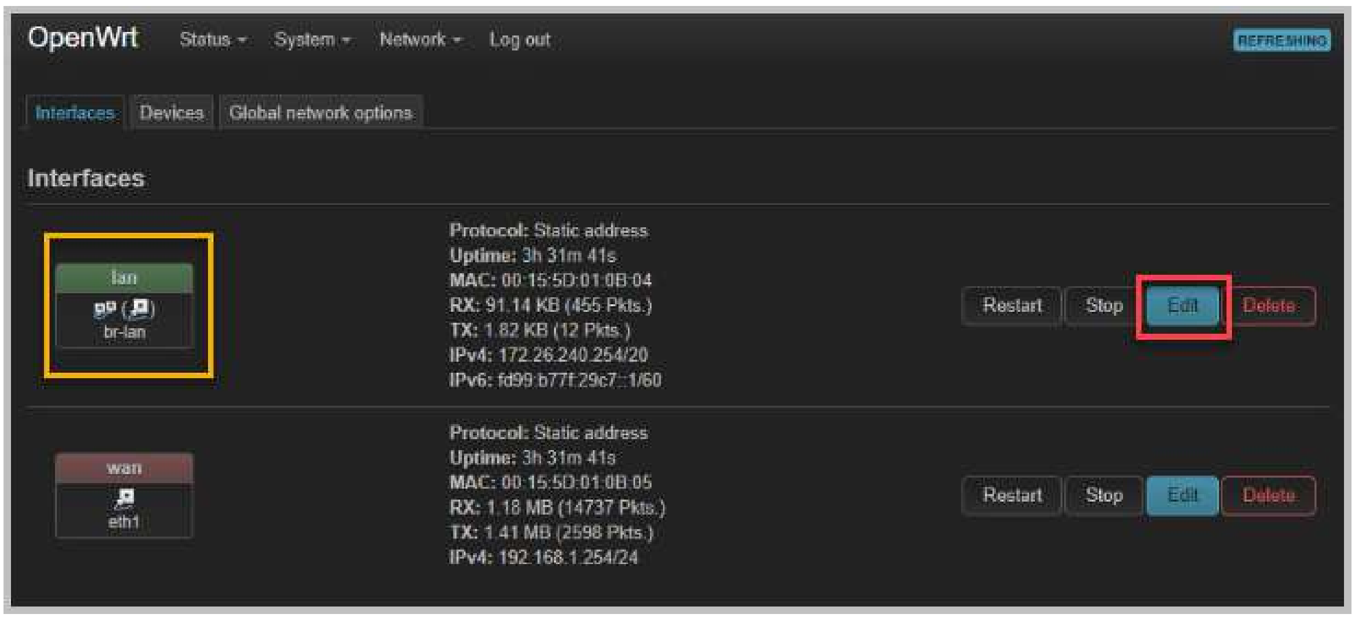
12. Leave the *Win11-OS OpenWRT* window open for the next lab activity.

2.6 Disabling Unused Ports

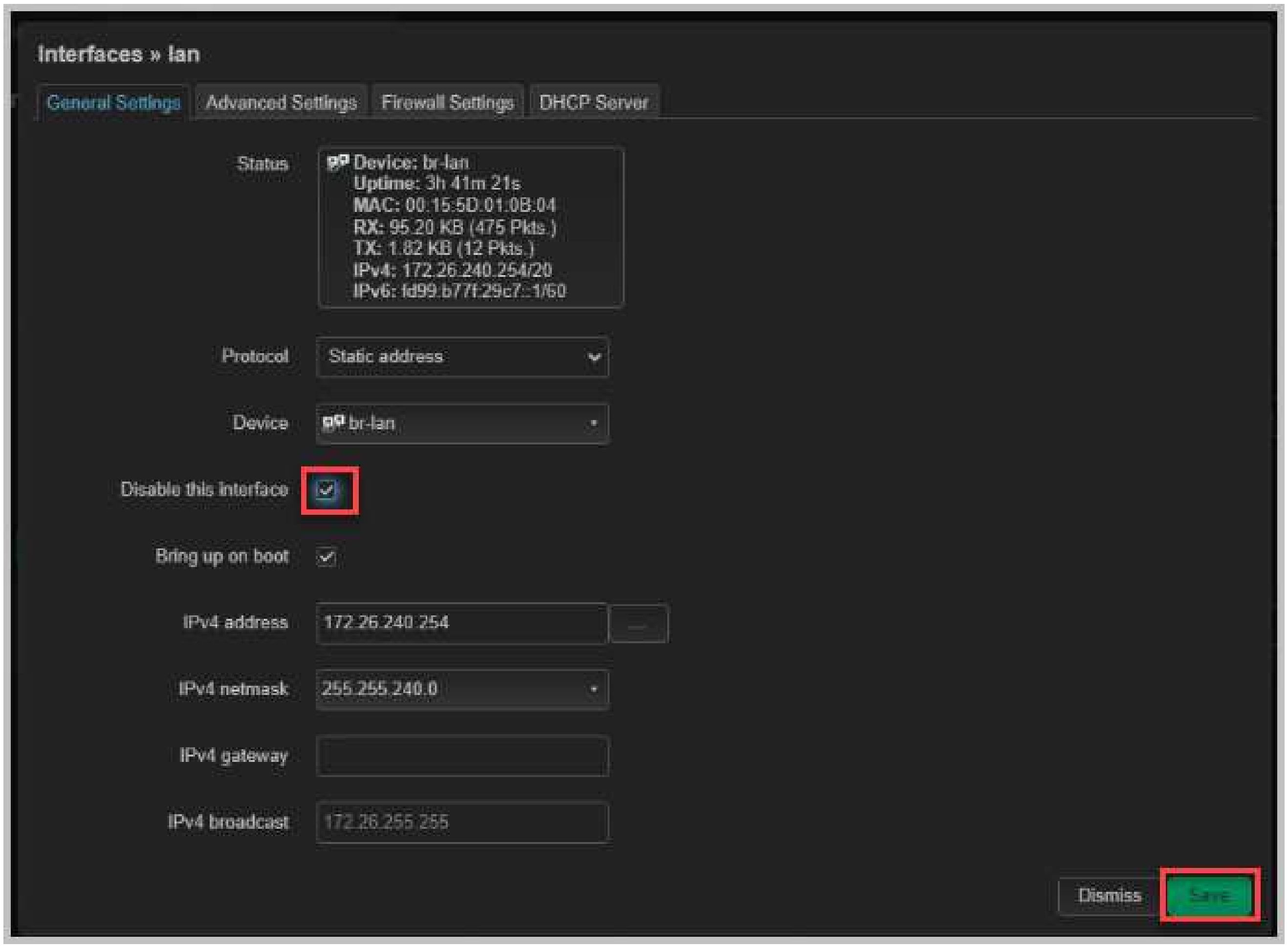
Disabling unused ports on your OpenWRT router is a critical measure that reduces the attack surface, enhances network performance, and improves privacy. By limiting the number of accessible services, you minimize potential vulnerabilities and optimize resource utilization. To implement this, identify unused ports, disable them using the LUCI interface or command-line tools, and regularly review your configuration to ensure only necessary ports remain active. Keeping your OpenWRT firmware up-to-date further strengthens your security posture.

1. In the *OpenWRT Luci* web interface, select **Network** and then **Interfaces.**

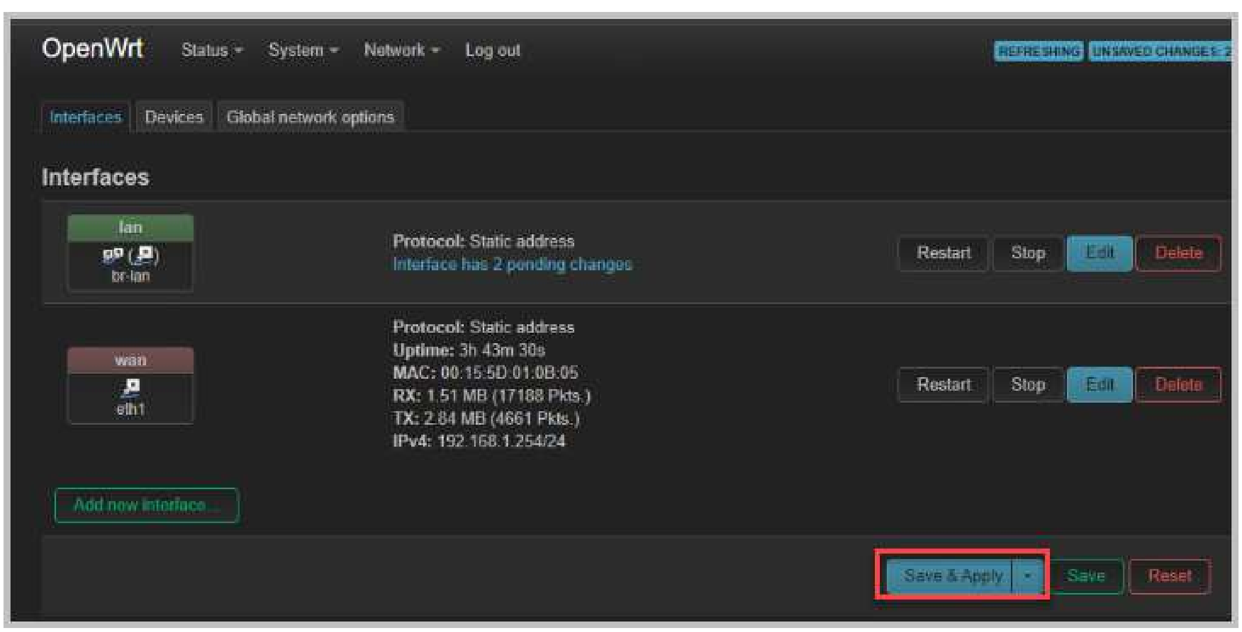
2. Review the list of active interfaces and identify which ones are not in use or are unnecessary for your current setup. For this lab, the **lan interface** does not have connectivity to the *Linux-OS or Win10-OS systems.* Thus, this interface needs to be disabled. Select **Edit** for the *lan interface.*



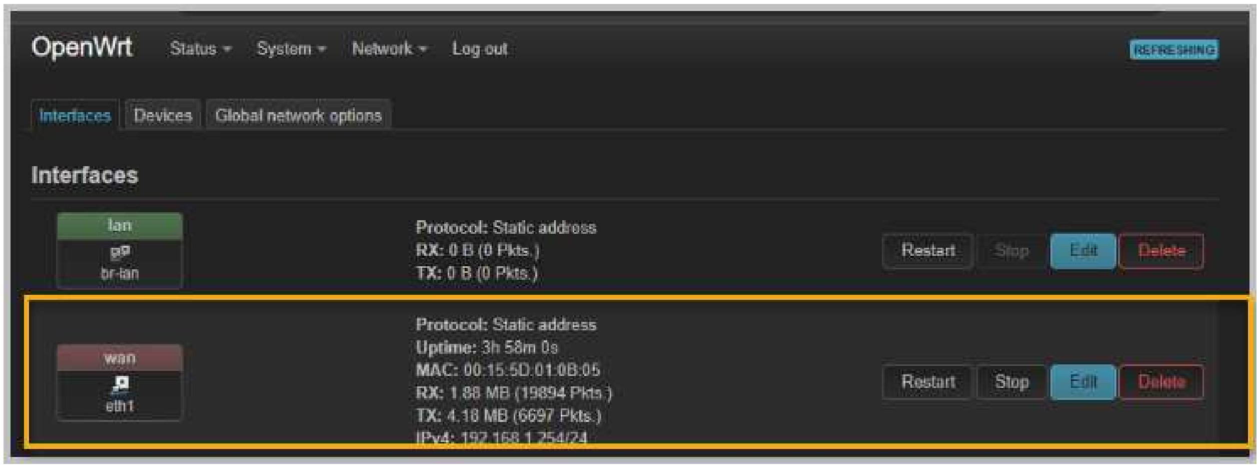
3. Select the **General Settings** tab, and place a **check mark** next to *Disable this interface.* Select **Save** to process the change.



4. Select **Save & Apply.** In OpenWRT, the "Save & Apply" button is used to commit any changes you've made to the router's configuration settings.



5. The only *OpenWRT Interface* that should be active is the **wan interface.**

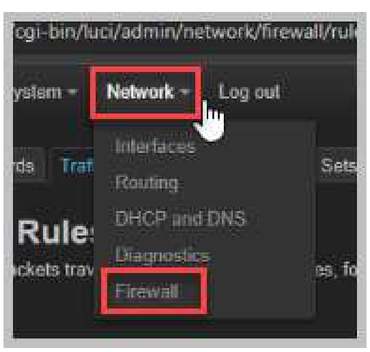


6. Leave the *Win11-OS System* open for the next activity.

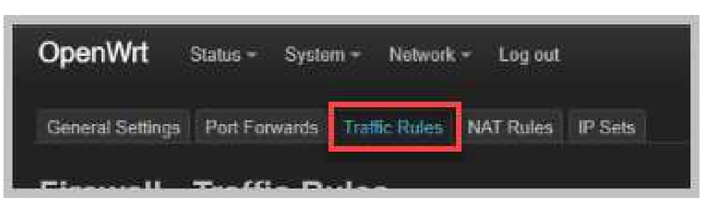
2.7 Set Content Filtering

Content filtering on *OpenWRT* offers numerous benefits, including enhanced security by blocking malicious websites and inappropriate content, improved network performance by reducing bandwidth consumption and speeding up page load times, increased privacy by limiting data exposure and blocking tracking scripts, and enforced productivity by restricting access to time-wasting websites and encouraging a more focused online experience. As this lab reservation cannot connect to outside websites such as www.cheese.com, the only option is to set content filtering for the address 192.168.1.254, which is the *IP Address* for *OpenWRT.*

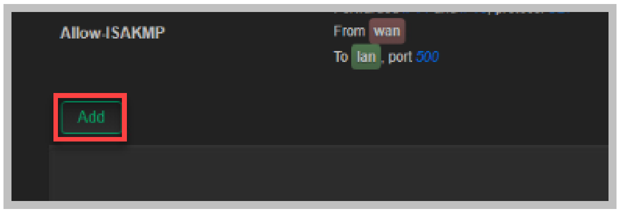
1. In the *OpenWRT LuCI* web interface, select **Network** and then **Firewall.**



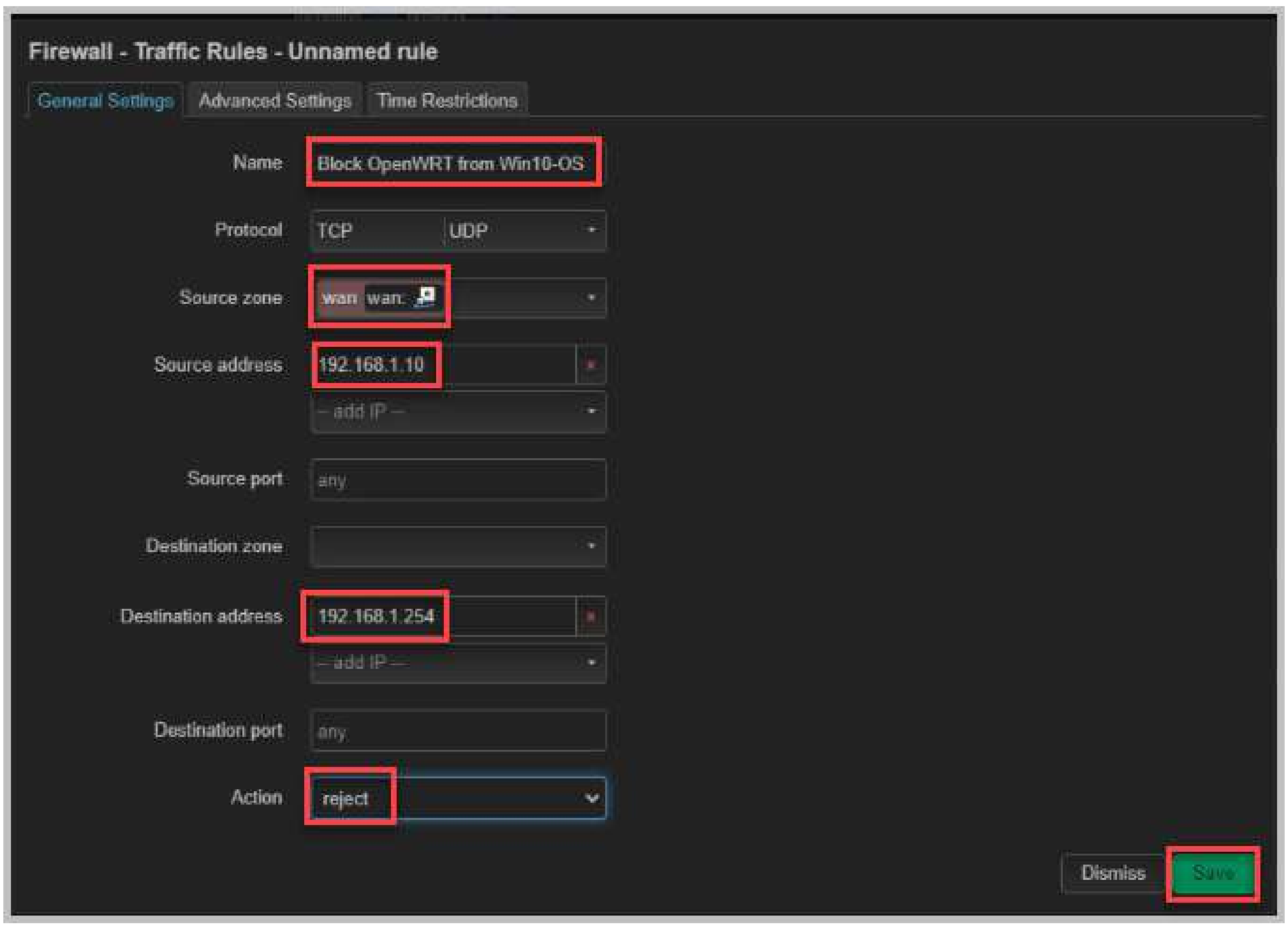
2. Select the **Traffic Rules** tab.



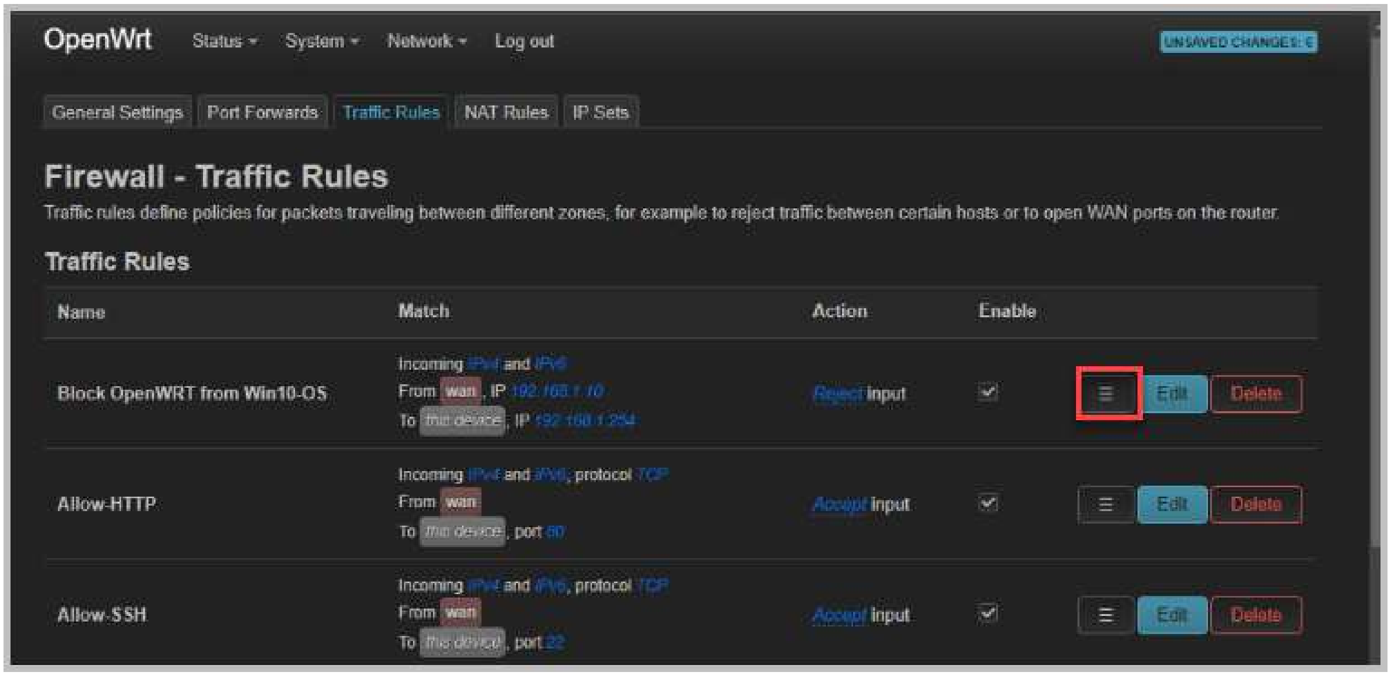
3. Scroll to the bottom of the *Traffic Rules Page,* and select **Add.**



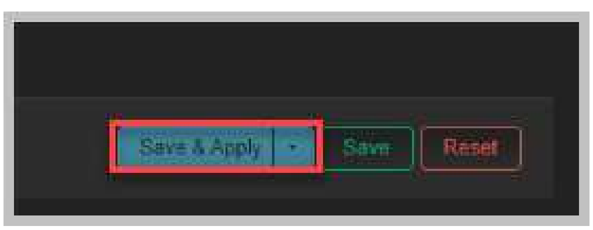
4. On the **General Settings** Tab, type **Block OpenWRT from Win10-OS** in the Name textbox, and change the Source zone to **Wan.** This sets a name for the Traffic Rule and isolates the WAN Zone Traffic by selecting the Wan Source zone. For the Source Address, select the IP Address for Win10-OS, which is **192.168.1.10.** For the Destination Address, select the IP Address for *OpenWRT,* which is **192.168.1.254.** For action, select **reject.** This sets the Firewall Traffic Rule to reject *Win10-OS* from connecting to *OpenWRT.* Select **Save.**



5. To ensure that the new rule is processed first and takes precedence over other rules, you'll need to move it to the top of the traffic rules list. By moving the new rules to the top of the list, you're ensuring that it's processed first by the firewall, effectively blocking access from *Win10-OS to OpenWRT.*
6. Move the new Traffic Rule to the beginning of the Traffic Rules List. First, move the **Block OpenWRT from Win10-OS** rule by selecting and holding down the **Drag to reorder** button and dragging it to the top of the list. Your Traffic Rules list should look exactly like the screenshot below.

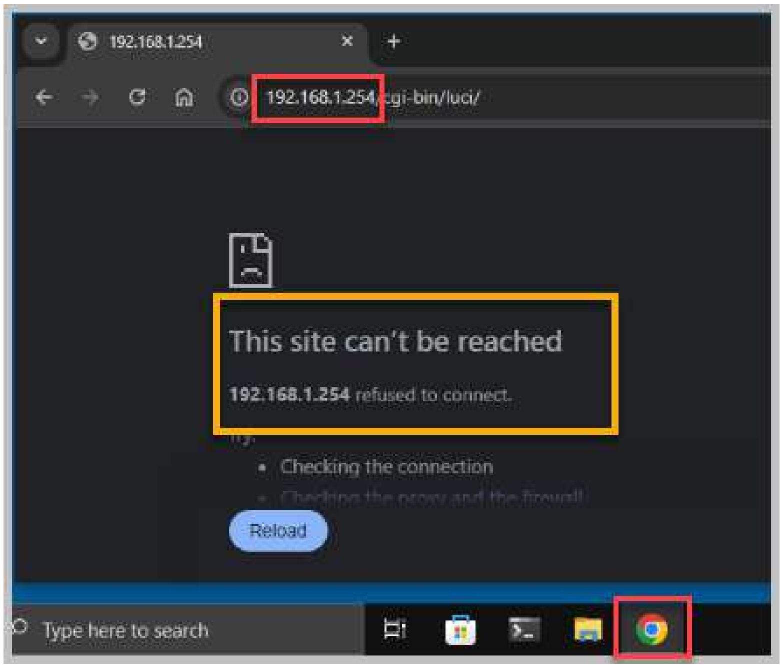


7. Scroll to the bottom of the *Traffic Rules* list, and click the **Save & Apply** button. This action will save your changes to the device's storage and immediately apply them to the active configuration. This ensures that the newly configured rules are enforced by the firewall.



8. Open a console connection to the **Win10-OS** system.

9. Select the **Google Chrome** icon on the taskbar. Enter **192.168.1.254** in the address window and select **Enter** on your keyboard. You should get a notification that *This site can't be reached.* This indicates that the new firewall traffic rule is working as intended.



10. The lab is now complete; you may end the reservation.